

(3 Hours)

[Total Marks : 80

- N. B. :** (1) Question No. 1 is **compulsory**.
 (2) Attempt any **three** from remaining.

1. (a) Write a recursive method to calculate factorial of a integer number. 5
 (b) Explain Bit-wise operators available in JAVA with example. 5
 (c) Explain how JAVA is platform - independent. 5
 (d) Write a program to find the largest of the three integers accepted from command line. 5

2. (a) Write a detailed note on System.arraycopy(). 5
 (b) Write a program to display following pattern :- 5
 1
 0 1
 1 0 1
 0 1 0 1
 (c) Explain different types of inheritance with example. 10

3. (a) With proper example explain the steps to create a package and add a class or interface. 5
 (b) Explain Thread life cycle. 5
 (c) Write a program in JAVA to accept the values of a, b, c and d. 10
 Calculate and display $((a*d) + (b*c)) / (b*d)$.
 Create user defined exception to display proper message when value of (b*d) is ZERO.

4. (a) What is Vector ? Explain any five methods of Vector. 5
 (b) Explain what are abstract class and abstract methods. 5
 (c) Write a program to find largest and second largest element of an array. 10

5. (a) Explain method overloading and method overriding with suitable example. 10
 (b) Write a program to accept a number from command line and to check whether it is Armstrong Number or Not. 10

6. Write short notes on (any four) :- 20
 (a) Wrapper classes.
 (b) Life cycle of Applet
 (c) Static members
 (d) Thread synchronization
 (e) Exception handling mechanism.